STRESS ZOMBIE APOCALYPSE

The Solid Mechanics Survival Engine • Complete Game Manual & Theoretical Guide

Core Focus: Mechanics of Solids • 2D Stress Transformation & Failure Criteria
Project Scope: Interactive 2D Stress Transformation, Mohr's Circle Geometry, Angle Doubling (2θ), Failure Theories & Hydrostatic Quenching
Technology Stack: HTML5 Canvas + Vanilla JavaScript + Web Audio Procedural Synthesis + Python 3 Daemon (Zero External Dependencies)

---

1. Executive Summary & Narrative Premise

In a dystopian bio-mechanical catastrophe, pathogen-altered zombies attack defense fortifications. Unlike organic creatures, these monstrosities are reinforced by an internal **plane stress tensor** comprising normal stresses (σ_x, σ_y) and shear stress (τ_{xy}). Conventional kinetic weapons (random bullets, blunt blades) glance harmlessly off their skin because their lattice bonds only fail when tensile or shear stress exceeds critical limits along **exact plane orientations (θ)**. As the lead defense engineer armed with a **Rotating High-Frequency Plane-Cutter Cannon**, you must calculate and dial the correct cutting angle θ to cleave the zombies along their fatal principal or maximum shear planes.

> **The Core Educational Breakthrough (Visualizing 2θ vs θ):**
> In solid mechanics, students routinely struggle to grasp why rotating a physical cutting plane by an angle θ causes the corresponding state of stress on **Mohr's Circle** to sweep through double the angle **2θ**. *Stress Zombie Apocalypse* transforms this abstract mathematical construct into a direct tactile experience: turning the blade dial by θ sweeps the live Mohr Radar pointer by 2θ. Students visually correlate physical blade orientation with circular stress coordinates in real time.

2. Target Audience & Target Users

This game was designed specifically for engineering education, active learning, and demonstration:

Target User Group	Primary Learning & Practical Application
Undergraduate Engineers (Mech, Civil, Aero, Materials)	Mastering 2D plane stress transformation formulas, identifying principal stresses (σ_1, σ_2), calculating principal angles (θ_p), and understanding maximum in-plane shear (τ_{max}).
Competitive Exam Candidates (GATE ME/CE, FE, ESE)	Rapid mental calculation of Mohr's circle center $C = (\sigma_x+\sigma_y)/2$, radius $R = \sqrt{[((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^2]}$, stress invariants I_1, and failure theories.
University Professors & TAs (Classroom Demos & Labs)	Interactive lecture visualizer; utilizes real-time Mohr's circle radar and the **Study Lab** sandbox for custom classroom demonstrations and problem solving.
STEM Gamers & Hackathons (Serious Games)	Engaging action-puzzle gameplay with responsive audio, dynamic slicing physics (upper torso slides along cut angle), screen shake, and progressive wave difficulty.

3. Master Controls Guide: What Every Key Does & Why

Every control input in *Stress Zombie Apocalypse* maps directly to a physical variable in continuum mechanics. The table below details every key, its operational function, and the exact physical rationale behind its use:

Key / Input	Mechanical Action	Why We Use It & Solid Mechanics Meaning
[A] Key or Left Arrow	Rotates Blade **Clockwise** ($-\Delta\theta$ by 1.5° / 5°)	**Physical Plane Rotation:** Decreases the cut angle θ relative to the positive horizontal axis. Shifts normal stress σ_θ and shear τ_θ along their sinusoidal transformation trajectories. Use [A] when the target failure angle is lower than the current blade orientation.
[D] Key or Right Arrow	Rotates Blade **Counter-Clockwise** ($+\Delta\theta$ by 1.5° / 5°)	**Standard Mechanics Convention:** In 2D continuum mechanics, counter-clockwise angular displacement from the +x axis is positive ($+\theta$). As you press [D], the physical blade rotates by θ, while the red radial pointer on Mohr's circle radar sweeps forward by **2**θ.
[Spacebar] or FIRE Button	**Fires High-Frequency Blade Cleave** along angle θ	**Strict Failure Threshold:** Spacebar triggers an instantaneous plasma cut. Unlike arcade games, **it does NOT deal damage at random angles**! If normal or shear stress on that plane is below the failure threshold, the shot deflects harmlessly with 0 damage. Only an aligned cut triggers critical fracture!
Rotary Dial or Angle Slider	Direct 360° Tactile Angle Positioning	Allows continuous mouse/touch dragging from 0° to 180°. Displays active blade angle θ in cyan and the corresponding Mohr circle angle 2θ in amber.
Snap Buttons (0°, 22.5°, 45°...)	Quick Failure Plane Jump Buttons	Instantly dials common textbook angles: 0° (uniaxial normal), 22.5° (Worked Example #1 principal plane), 45° (pure shear plane), and 67.5° (maximum shear plane).
[1] Slow-Mo	Time Dilation to 20% (5 sec)	Slows zombie approach so students have time to calculate θ_p on scratch paper or verify Mohr coordinates before firing.
[2] Hint	Holographic Target Guide	Projects a bright green dashed guide line directly onto the dial and canvas, illuminating the exact optimal angle θ_{opt}.
[3] Scanner	Stress Tensor Diagnostic	Decodes hidden stress components in Hard (Exam) mode, revealing numerical values of σ_x, σ_y, and τ_{xy}.
[4] I_1 Shield	First Invariant Energy Barrier	Harnesses the principle of rotational invariance: $I_1 = \sigma_x + \sigma_y = \sigma_1 + \sigma_2$ = Constant. Generates a 4-second invulnerability shield that absorbs all incoming kinetic impact.
[5] Hydro-Jet	Hydrostatic Deluge Cannon	**Pure Hydrostatic Compression:** Blasts enemies with omnidirectional fluid pressure $\sigma_m = (\sigma_x+\sigma_y)/2$. This eliminates in-plane shear ($\tau \to 0$), collapses Mohr's circle radius to zero ($R = 0$), and quenches ductile zombies into brittle glass!

4. Why Does Spacebar Only Work on Specific Angles?

A common question from new players is: *'Why did my shot bounce off when I pressed Spacebar?'*
The Mechanical Reality: If a brittle material has a tensile fracture limit of σ_Y = 92.4 MPa, but the player cuts at an angle where normal stress is only σ_θ = 20.0 MPa, the tensile stress across that plane is insufficient to break internal molecular bonds. The game enforces realistic failure physics:

- **Off-Angle Shot ($|\theta - \theta_{opt}| > 7.5°$):** The blade glances off with a loud metallic *clang*. Damage is 0. An alert informs the player: *'DEFLECTED! Stress is only 20 MPa. Need 92.4 MPa at θ_p = 22.5°!'*
- **Aligned Shot ($|\theta - \theta_{opt}| \leq 7.5°$):** The on-canvas reticle turns luminous green with a **LOCK** badge. Pressing Spacebar delivers a **CRITICAL CLEAVE**, instantly severing the zombie while its upper torso physically slides along angle θ.

5. The Solid Mechanics Engine: Mathematical Derivations

All real-time gameplay calculations, HUD radar sweeps, and curve plots run on exact analytical solutions from continuum mechanics:

1. Normal Stress on Inclined Plane θ:	$\sigma_\theta = (\sigma_x + \sigma_y)/2 + [(\sigma_x - \sigma_y)/2]\cdot\cos(2\theta) + \tau_{xy}\cdot\sin(2\theta)$
2. Shear Stress on Inclined Plane θ:	$\tau_\theta = -[(\sigma_x - \sigma_y)/2]\cdot\sin(2\theta) + \tau_{xy}\cdot\cos(2\theta)$
3. Mohr's Circle Center C:	$C = (\sigma_x + \sigma_y) / 2$ (Mean Normal Stress σ_m)
4. Mohr's Circle Radius R:	$R = \sqrt{[\ ((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\ ]}$
5. Principal Stresses ($\tau = 0$):	$\sigma_1 = C + R$ (Maximum Tensile), $\sigma_2 = C - R$ (Minimum Principal)
6. Principal Plane Angle θ_p:	$\tan(2\theta_p) = 2\tau_{xy} / (\sigma_x - \sigma_y) \Rightarrow \theta_p = \frac{1}{2}\ \text{atan2}(2\tau_{xy}, \sigma_x - \sigma_y)$
7. Maximum Shear Stress & Planes:	$\tau_{max} = R = (\sigma_1 - \sigma_2)/2$ occurring at $\theta_s = \theta_p \pm 45°$
8. First Stress Invariant I_1:	$I_1 = \sigma_x + \sigma_y = \sigma_1 + \sigma_2 = \sigma_\theta + \sigma_{\theta+90°}$ = Constant (Rotational Invariance)
9. Second Stress Invariant I_2:	$I_2 = \sigma_x\sigma_y - \tau_{xy}^2 = \sigma_1\sigma_2$ = Constant
10. von Mises Equivalent Stress σ_{vm}:	$\sigma_{vm} = \sqrt{[\ \sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2\ ]} = \sqrt{[\ \sigma_x^2 - \sigma_x\sigma_y + \sigma_y^2 + 3\tau_{xy}^2\ ]}$
11. Hydrostatic Stress Component:	$\sigma_{hydro} = (\sigma_x + \sigma_y)/2$; Deviatoric tensor: $s_x = \sigma_x - \sigma_{hydro}$, $s_y = \sigma_y - \sigma_{hydro}$
12. Generalized Hooke's Law (Strains):	$\varepsilon_x = (\sigma_x - \nu\sigma_y)/E$, $\varepsilon_y = (\sigma_y - \nu\sigma_x)/E$, $\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$
13. Engineering Shear Strain γ_{xy}:	$\gamma_{xy} = \tau_{xy} / G$ where $G = E / [2(1 + \nu)]$; Tensorial shear $\varepsilon_{xy} = \gamma_{xy}/2$
14. Volumetric Dilatation e & Energies:	$e = \varepsilon_x + \varepsilon_y + \varepsilon_z = [(1-2\nu)/E]I_1$ • Total Energy: $u = \frac{1}{2}(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \tau_{xy}\gamma_{xy})$ Distortion Energy: $u_d = [(1+\nu)/(3E)]\sigma_{vm}^2$

6. Step-by-Step Solved Worked Example (Zombie #1 from Lecture 4)

The following numerical problem is solved live inside the game engine and unit-tested in test_mechanics.py:

Step	Formula & Exact Numerical Substitution	Resulting Output
Given Tensor	Direct normal and shear stresses on the initial horizontal/vertical element	**$\sigma_x = 80$ MPa, $\sigma_y = 20$ MPa, $\tau_{xy} = 30$ MPa**
1. Center C	$C = (\sigma_x + \sigma_y)/2 = (80 + 20) / 2$	**C = 50.0 MPa**
2. Radius R	$R = \sqrt{[\ ((80 - 20)/2)^2 + 30^2\ ]} = \sqrt{[\ 30^2 + 30^2\ ]} = \sqrt{1800}$	**R = 42.43 MPa**
3. Principal Stresses	$\sigma_1 = C + R = 50 + 42.43$ $\sigma_2 = C - R = 50 - 42.43$	**$\sigma_1 = 92.43$ MPa (Max Tensile)** **$\sigma_2 = 7.57$ MPa (Min Tensile)**
4. Principal Angle θ_p	$\tan(2\theta_p) = 2(30) / (80 - 20) = 60/60 = 1.0 \Rightarrow 2\theta_p = 45°$	**$\theta_p = 22.5°$ (Cutting Angle)**
5. Max Shear & Plane	$\tau_{max} = R = 42.43$ MPa at $\theta_s = \theta_p + 45° = 22.5° + 45°$	**$\tau_{max} = 42.43$ MPa at 67.5°**
Cut at $\theta = 22.5°$ (Target Cleave)	$\sigma_{22.5} = 50 + 30\cdot\cos(45°) + 30\cdot\sin(45°) = 50 + 21.21 + 21.21$ $\tau_{22.5} = -30\cdot\sin(45°) + 30\cdot\cos(45°) = 0.0$ MPa	**$\sigma_\theta = 92.43$ MPa • $\tau = 0$** **CRITICAL CLEAVE SHATTERS ZOMBIE!**
Cut at $\theta = 90.0°$ (Misaligned)	$\sigma_{90} = 50 + 30\cdot\cos(180°) + 30\cdot\sin(180°) = 50 - 30 + 0$ $\tau_{90} = -30\cdot\sin(180°) + 30\cdot\cos(180°) = -30.0$ MPa	**$\sigma_\theta = 20.0$ MPa (<< 92.4 MPa)** **BLADE DEFLECTS (0 DAMAGE)**
6. Hooke's Law Strains (E=200 GPa, ν=0.30)	$\varepsilon_x = (80 - 0.3(20))/200000 = 74/200000 = 370\ \mu\varepsilon$ $\varepsilon_y = (20 - 0.3(80))/200000 = -4/200000 = -20\ \mu\varepsilon$ $\varepsilon_z = -0.3(100)/200000 = -150\ \mu\varepsilon$	**$\varepsilon_x = 370.0\ \mu\varepsilon$, $\varepsilon_y = -20.0\ \mu\varepsilon$** **$\varepsilon_z = -150.0\ \mu\varepsilon$**
7. Shear Strain & Dilatation	$G = 200000/[2(1.3)] = 76923$ MPa $\gamma_{xy} = 30/76923 = 390.0\ \mu$rad Dilatation $e = 370 - 20 - 150 = 200.0\ \mu\varepsilon$	**$\gamma_{xy} = 390.0\ \mu$rad** **$e = 200.0\ \mu\varepsilon$ (Volumetric Expansion)**
8. Strain Energy Density	$u = \frac{1}{2}[80(370) + 20(-20) + 30(390)]\times10^{-3} = 20.45$ kJ/m³ $u_d = [(1.3)/(600000)](7900)\times1000 = 17.12$ kJ/m³	**$u_{total} = 20.45$ kJ/m³** **$u_{distortion} = 17.12$ kJ/m³ (83.7%)**

7. Zombie Bestiary & Failure Criteria Mapping (Lectures 31-32)

Each enemy class directly represents a classical engineering material failure theory. Players must observe the zombie's visual appearance and aura to select the matching failure plane:

Zombie Class	Failure Criterion & Formula	Tactical Weakness & Target Cutting Plane
Brittle Biter (Crystalline Blue / Ice Glass)	**Rankine Theory** (Maximum Normal Stress) $\sigma_1 \geq \sigma_Y$	**Tensile Cleave Plane:** Brittle materials (glass, concrete, cast iron) fail via cleavage under peak tensile stress with zero plastic deformation. Align blade with principal plane $\theta = \theta_p$ where shear stress vanishes ($\tau = 0$).
Ductile Dragger (Amber Neon / Molten Metal)	**Tresca Theory** (Maximum Shear Stress) $\tau_{max} \geq \sigma_Y / 2$	**Shear Slip Plane:** Ductile materials (mild steel, aluminum) yield along 45° slip planes governed by peak shear stress. Align blade with maximum shear planes: $\theta_s = \theta_p \pm 45°$ (the top/bottom poles of Mohr's circle radar).
Shapeshifter (Pulsing Magenta Matrix)	**Dynamic Rankine ↔ Tresca** (Alternating State Tensor)	**Adaptive Targeting:** Mutates its internal stress tensor during movement! When pulsing cyan, strike its principal plane θ_p; when glowing orange, instantly pivot the dial by 45° to strike its shear plane θ_s.
Twin Zombies (Linked Cyan Pair)	**Coupled Plane Cleave** (Tensor Harmonics)	Two zombies marching in tandem with coupled stress tensors. Precision alignment allows one cleave cut to slice through both bodies simultaneously for a massive score multiplier.
von Mises Mutant (Colossal Wave 5 Boss)	**Distortion Energy Theory** $\sigma_{vm} \geq \sigma_Y$	**Energy Maximization:** Massive titanium-reinforced boss. Withstands pure tensile or pure shear cuts. Requires a balanced cut angle that maximizes effective von Mises distortion energy: $\sigma_{eff} = \sqrt{(\sigma_\theta^2 + 3\tau_\theta^2)}$.

8. The In-Game Diagnostic Tools & Visual Hints

To empower students to succeed without memorizing formulas, the game provides 6 visual instruments and real-time hints:

Visual Diagnostic Tool	On-Screen Display & Real-Time Functionality
1. Live Dial Direction Arrow (Interactive Rotary Dial)	Displays an intuitive direction indicator right above the dial: "PRESS [D] → (15° AWAY)" or "← PRESS [A] (8° AWAY)". Tells the player exactly which key to press and how far to turn.
2. Target Reticle & Guide Line (Battlefield Viewport)	A laser tracking line connects the turret cannon to the nearest zombie. Over the zombie's body, a green dashed failure line indicates θ_{opt}. When the blade angle enters ±7.5°, the reticle locks on and glows green with LOCK.
3. Mohr's Circle Radar (Top-Right Canvas)	Visualizes the complete Mohr circle with center C and radius R. Plots reference points X(σ_x, τ_{xy}) and Y(σ_y, $-\tau_{xy}$). A bright red pointer rotates by **2**θ, while green ticks highlight failure points σ_1 and τ_{max}.
4. Transformation Curves (Figure 2 Cartesian Graph)	Plots σ_θ (cyan wave) and $\lvert\tau_\theta\rvert$ (amber wave) from 0° to 180°. A vertical tracking needle indicates the player's active blade angle relative to the mathematical peaks.
5. Floating Stress Element (Above Every Zombie)	A floating infinitesimal square element with color-coded stress arrows: red horizontal vectors for σ_x, green vertical vectors for σ_y, and orange shear vectors for τ_{xy}. The element rotates live as the zombie moves.
6. Real-Time Deflection HUD (Combat Feedback Box)	When an off-angle cut is made, the HUD flashes red and prints the exact physics failure breakdown: "DEFLECTED! Current normal stress is only 20 MPa. Need 92.4 MPa at θp = 22.5°!"

9. Step-by-Step Gameplay Guide: How to Play

A typical combat round follows a structured 7-step engineering decision workflow:

Step	Player Action	Underlying Solid Mechanics Decision
Step 1: Inspect	Observe the incoming zombie type and aura color.	Identifies failure theory: Brittle (Ice Blue) → Rankine; Ductile (Amber) → Tresca; Boss → von Mises.
Step 2: Read Tensor	Read σ_x, σ_y, τ_{xy} from the floating element or HUD.	Extracts given stress state components in MegaPascals (MPa).
Step 3: Solve Plane	Calculate target angle $\theta_p = ½ \text{ atan2}(2\tau_{xy}, \sigma_x - \sigma_y)$.	For Brittle: target $\theta_{opt} = \theta_p$. For Ductile: target $\theta_{opt} = \theta_p \pm 45°$.
Step 4: Rotate Blade	Press [A] (clockwise) or [D] (counter-clockwise), or drag dial.	Transforms the cutting blade through physical angle θ; sweeps Mohr's circle radar by 2θ.
Step 5: Confirm Lock	Watch the aiming reticle turn luminous green.	Confirms that $\lvert\theta - \theta_{opt}\rvert \le 7.5°$ and transformed stress reaches critical threshold.
Step 6: Cleave!	Press [Spacebar] or click FIRE button.	Fires plane-cutter laser; zombie upper body slides off along angle θ with 3x Critical Shatter!
Step 7: Power-Ups	Deploy keys [1] to [5] when overwhelmed.	Use Hydro-Jet [5] to quench ductile zombies into brittle targets, or I_1 Shield [4] to deflect attacks.

10. The 10-Sector Campaign: Detailed Level Progression & Locking Matrix

The campaign consists of 10 pedagogical combat sectors. Advanced sectors remain locked until preceding sectors are cleared:

Sector / Level	Stress Tensor State	Primary Failure Theory	Pedagogical Mechanics Focus
Level 1: Tension Outpost	$\sigma_x = 80\text{-}120$, $\sigma_y = 0$, $\tau_{xy} = 0$	Rankine (Tensile Cleavage)	Uniaxial tension with zero shear. Physical cuts align directly at $\theta = 0°$ and 90°.
Level 2: Compression Catacombs	$\sigma_x = 80\text{-}110$, $\sigma_y = 20\text{-}40$, $\tau = 0$	Rankine (Biaxial State)	Biaxial normal stresses. Demonstrates Mohr center shift to $C = (\sigma_x + \sigma_y)/2$.
Level 3: Shear Storm	$\sigma_x = 60\text{-}90$, $\sigma_y = 10\text{-}20$, $\tau_{xy} = \pm 35$	Rankine on Rotated Axes	In-plane shear rotates principal axes! Calculates $\theta_p = ½ \text{ atan2}(2\tau_{xy}, \sigma_x - \sigma_y)$.
Level 4: Ductile Threat	$\sigma_x = 80$, $\sigma_y = 20$, $\tau_{xy} = 30$	Rankine vs. Tresca Yield	Switching between normal cleavage (θ_p) and maximum shear slip planes ($\theta_s = \theta_p \pm 45°$).
Level 5: Hydro-Quench Trench	Heavy ductile metal armor	Hydrostatic Fluid Quenching	Hydro-Jet [Key 5] wipes deviatoric shear to $\tau = 0$, collapsing Mohr circle to $R = 0$ into brittle glass.
Level 6: Quantum Shapeshifters	Dynamic tensor mutations	Adaptive Rankine ↔ Tresca	Rapid cognitive switching between principal plane and 45° shear plane as aura shifts.
Level 7: Twin Resonance	Coupled tensor harmonics	Coupled Plane Cleaving	Synchronized blade timing to cleave two coupled entities in one simultaneous stroke.
Level 8: Invariant Citadel	High-speed swarm assault	First Stress Invariant I_1	Deploys First Invariant Shield [Key 4] ($I_1 = \sigma_x + \sigma_y = \text{const}$) for kinetic defense.
Level 9: von Mises Titan Boss	Colossal boss (600 HP)	Distortion Energy Theory	Delivers multi-axial energy cleaves: $\sqrt{(\sigma_\theta^2 + 3\tau_\theta^2)} \ge \sigma_Y$.
Level 10: Endless Void	Randomized multi-axial tensors	All Failure Theories	Infinite survival mode unlocked after conquering Campaign Levels 1 through 9.

11. The Interactive Study Lab & Derivation Sandbox

Clicking the **'STUDY LAB'** button on the top navigation bar opens an interactive, non-combat educational laboratory. Students can:

- **Input Any Custom Stress State:** Sliders allow adjusting σ_x (–100 to +100 MPa), σ_y, τ_{xy}, and θ.
- **Inspect Rotated Infinitesimal Element:** Displays an isolated square element rotating in real time, showing exact normal and shear tractions acting on all four faces.
- **Load Textbook Presets:** Instant buttons for *Worked Example #1*, *Pure Shear*, *Uniaxial Tension*, and *Hydrostatic State*.
- **Read Step-by-Step Derivations:** Formatted LaTeX-style formulas show numerical values substituting into each equation.

12. Post-Level Study Lab Forensics & Level Map Navigation Flow

Campaign navigation follows a structured loop: **Welcome Screen** → **Level Selection Map** → **Sector Combat** → **Post-Level Stress & Strain Lab Report** → **Level Selection Map**. During active wave combat, zombies cleave cleanly without interrupting popup reports. Instead, upon clearing each level, the **Post-Level Forensic Stress & Strain Lab** opens automatically with the interactive layout of the Study Lab:

• **Interactive Specimen Tabs & Canvases:** Click any cleaved zombie from the completed sector to inspect its live **Mohr's Circle** and **Rotated Stress & Strain Element** with an interactive exploration angle slider θ.

• **Full Stress Tensor Matrix [σ]:** Displays [σ_x, τ_{xy}; τ_{yx}, σ_y], Mohr center C, radius R (τ_{max}), principal stresses σ_1, σ_2, invariants I_1, I_2, and von Mises stress σ_{vm}.

• **Full Strain Tensor Matrix [ε]:** Evaluates 3D/2D strains via Generalized Hooke's law [ε_x, ε_y], Poisson out-of-plane normal strain ε_z, engineering shear strain γ_{xy}, principal strains ε_1, ε_2, γ_{max}, and volumetric dilatation e.

• **Strain Energy Density Partitioning & Return to Level Map:** Shows total strain energy density u, volumetric energy u_v, distortion energy u_d, and complete mathematical derivations; closing the report unlocks the next sector and returns cleanly to the **Level Selection Map**.

13. How to Run, Test & Deploy the Project

• **Local Server Execution:** Run python run_game.py in the project directory. The multi-threaded server immediately opens **http://localhost:8000/** in your default web browser.

• **Automated Unit Testing:** Run python test_mechanics.py. Verifies **11/11** solid mechanics and strain tensor calculations against analytical standards (Pass rate: 100% in 0.000s).

• **Zero-Install Web Deployment:** Push to GitHub and enable **GitHub Pages**. The game runs directly in Google Chrome, Microsoft Edge, Safari, and Firefox with zero server installations, no npm packages, and zero external downloads.